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Early prediction of ventricular fibrillation through continuous ECG analysis and machine learning techniques

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Abstract

Purpose: Ventricular fibrillation (VF) is a life-threatening arrhythmia in which timely intervention is critical to prevent sudden cardiac death (SCD). Early prediction can improve survival rates by enabling pre-emptive clinical actions or early patient response. However, most existing VF prediction methods rely on isolated electrocardiogram (ECG) segments at fixed time points, limiting their practical application. This study introduces a continuous ECG analysis framework for early VF prediction using heart rate variability (HRV) features.

Methods: ECG data from the PhysioNet Sudden Cardiac Death (SCD) and Normal Sinus Rhythm (NSR) databases were analysed. HRV features, including time-domain and nonlinear metrics, were extracted from 91-minute windows preceding VF onset. A Gaussian kernel Support Vector Machine (SVM) was trained to classify pre-VF and normal rhythms, with performance assessed via leave-one-out cross-validation. Alarm generation was based on a 10-minute moving average of SVM outputs exceeding a defined threshold.

Results: The proposed algorithm achieved 100% sensitivity, 77.8% specificity, and 89.5% accuracy. The average earliest prediction time was 77.0 (±15.9) minutes, providing substantial lead time for clinical intervention.

Conclusion: This study demonstrates the feasibility of continuous ECG-based VF prediction, supporting its potential integration into clinical decision support systems and long-term monitoring platforms. Future work will focus on validation in wearable technologies and clinical workflows.

Keywords: Ventricular fibrillation (VF), Sudden cardiac death (SCD), Heart rate variability (HRV), Electrocardiogram (ECG), Support vector machine (SVM), Continuous prediction

1 Introduction

Sudden cardiac death (SCD) accounts for approximately 400,000 deaths annually in the United States, representing nearly half of all cardiac-related fatalities [1]. Sudden cardiac arrest (SCA), the unexpected and abrupt loss of heart function, can occur in individuals with or without a history of heart disease. SCD typically results from SCA and often

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happens within one hour of its symptom onset. Approximately 0.1% of the population in the United States and Europe experiences out-of-hospital SCA annually, as assessed by medical services [2]. Ventricular fibrillation (VF), a rapid and disorganised heart rhythm, is widely recognised as the most common underlying mechanism leading to SCA and hence SCD. Prompt medical intervention is vital when VF occurs, as the survival rate of the patient reduces by 10% for every minute of delay in resuscitation. Although up to three-quarters of people experience warning symptoms, such as chest discomfort, for a median of one hour before VF events, these symptoms are often overlooked or unrecognized [3]. Thus, automated prediction prior to VF events is critical to improving patient survival rates, enabling earlier attention to the problem and allowing more time for treatment preparation.

The ability to predict VF events using electrocardiogram (ECG) signals has significant potential for both in-hospital monitoring and ambulatory settings. In particular, early and reliable VF prediction could support deployment in wearable cardiac monitoring devices, such as patch-based ECG monitors, smartwatches, or implantable loop recorders [4], enabling continuous risk assessment beyond clinical environments. Numerous studies have attempted to predict future VF events using ECG signals and heart rate variability (HRV) analysis. These analyses require ECG or HRV data seconds, minutes, or hours before the events. In addition to conventional classification metrics such as accuracy, sensitivity, and specificity, prediction time serves as a crucial performance measure. Historically, this VF prediction problem has been framed as a classification task at a specific prediction time. This approach relied primarily on features extracted from isolated signal segments, categorising them as either pre-VF or control, and evaluating algorithm performance at the fixed prediction times. For example, [5] targeted a prediction time of three minutes by analysing only one-minute segments acquired four minutes before onset and achieved an accuracy of 94.74% at this specific prediction time.

Previous studies have sought to expand prediction windows and explore multi-interval prediction strategies [6]. For instance, an early study by [7] investigated multiple one-minute intervals at varying prediction times up to three hours before VF onset. This study achieved an accuracy of 80.6% at 50th minute before VF onset using time-domain HRV features. On the other hand, [8] reported an average accuracy of 90.18% over a 13-minute prediction window by employing time-local feature selection to manage different subsets of features tailored to each prediction time for higher classification accuracies. This methodology was further refined by [9], which achieved 97.6% accuracy 13 minutes before VF onset and 98.0% at 14th minute.

Recent research has pushed the boundaries of VF prediction, aiming to achieve both longer prediction windows and high accuracy. For example, [10] analysed multiple 10-minute HRV segments up to 60 minutes before VF onset and achieved an average prediction accuracy of 91.22% across the entire 70-minute window using multi-scale variation features combined with Shannon entropy and a time-domain feature. In [11], the authors targeted prediction times of 20, 30, 40, and 50 minutes before VF onset. They extracted 60 segments of 3 s randomly from each control recording and equivalent segments from SCD recordings prior to the designated prediction times. For each prediction time, 1080 normal sinus rhythm (NSR) ECG segments and 1200 pre-VF segments were used for feature extraction, model training, and testing. Employing an ensemble of

classifiers alongside time-domain, spectral, and complexity features, the study achieved over 99% accuracy for prediction times at 20, 40, and 50 minutes.

Another notable contribution is the work by [12], which extracted 31 one-minute segments from SCD recordings and one random segment from each control recording. The study applied complete ensemble empirical mode decomposition (CEEMD) for feature extraction and conducted statistical analysis on all signal segments up to 30 minutes before VF onsets. Using an SVM classifier, the model achieved over 90% accuracy for all minute-by-minute classifications, culminating in a 97.28% accuracy at the 30th minute. Singhal et al. [13] focused on a prediction time of at least 10 minutes. They extracted twenty one-minute signal segments 10 minutes before VF onset from SCD recordings and randomly selected segments from control recordings of subjects with normal sinus rhythm or congestive heart failure. By applying Fourier-based decomposition, look-ahead pattern extraction, principal component analysis and machine learning classification, they achieved an exceptional 100% accuracy for this prediction window.

These recent studies highlight significant advancements in VF prediction, leveraging sophisticated feature extraction methods and machine learning classifiers to extend prediction windows. However, many of these approaches rely on segment-based analysis and develop separate models for each prediction time. Although effective in achieving high accuracy, this strategy increases computational complexity and may limit real-world clinical application.

In contrast, a limited number of studies have adopted continuous analysis frameworks that evaluate signals in their entirety. For example, [14] analysed complete ECG recordings using eight-minute signal windows, achieving high accuracy (98.44%) and an average prediction time of 4 min 31 s. A follow-up study by [15] improved the phase space reconstruction technique and extended the evaluation across multiple databases, maintaining over 90% accuracy and a prediction time of more than four minutes. While they simulate continuous analysis, they fall short in achieving extended prediction times, lacking the integration of long prediction windows with high accuracy essential for clinical applications.

Based on the review, most existing VF prediction methods rely on isolated signal segments and focus on specific prediction time points. Little prior research has systematically implemented VF prediction algorithms in a way that supports continuous analysis and achieves an early prediction time, which is essential for clinical application. To address these limitations, this study introduces an improved VF prediction algorithm with the following contributions:

1. Continuous analysis: By analysing entire signal lengths, our approach captures evolving VF risk without preselecting specific segments, providing more robust and clinically relevant predictions.
2. High accuracy with extended prediction time: The algorithm achieves high predictive accuracy and early prediction times, offering greater lead time for timely and potentially life-saving interventions.

This study aims to bridge the gap between segment-based and continuous analysis frameworks, presenting a VF prediction algorithm that combines high accuracy, early

prediction times, and feasibility for practical implementation in clinical and wearable cardiac monitoring.

2 Methodology

2.1 Dataset

In this study, the ECG signals were sourced from the Physionet databases [16]. Pre-VF signals were obtained from the Sudden Cardiac Death Holter Database (SDDB) [17], while control signals were extracted from the MIT-BIH Normal Sinus Rhythm Database (NSRDB) [16]. The SDDB contains 23 long-term (24-hour) ECG recordings from individuals who experienced sustained ventricular arrhythmia; however, three records ('40', '42', '49') were excluded due to missing VF onset annotations. The NSRDB consists of 18 24-hour ECG records from individuals with normal sinus rhythm and no significant arrhythmia, all of which were included in our analysis. Only the first ECG channel from each subject's record was analysed to ensure independence between signal samples and prevent overly optimistic results. Since our study focused on HRV rather than raw ECG signals, any channel providing accurate R-peak detection was deemed suitable. In the SDDB, VF onset annotations were used, whereas in the NSRDB, a consistent reference point at the two-hour mark was used to avoid random signal selection, ensuring reproducibility of the test results.

2.2 Study workflow

Firstly, one-minute pre-VF and control signals were acquired from 20 SDDB records and 18 NSRDB records, respectively. A one-minute signal length was selected to ensure a fair comparison with the majority of previous predictive studies [9, 12, 13, 18]. Only the first ECG channel was extracted for each subject in SDDB and NSRDB. The VF onset annotations in SDDB and the two-hour time point in NSRDB were consistently used as reference points for signal acquisition. The one-minute ECG signals were then preprocessed using a fourth-order Butterworth filter with a 1 to 30 Hz passband. R-peak detection in the filtered signals was performed using the Hamilton-Tompkins algorithm to derive HRV signals, represented as time series of RR intervals. Subsequently, various time-domain and nonlinear HRV features were extracted from the derived HRV signals.

A previous study explored feature selection and model hyperparameters to develop a prediction algorithm with high accuracy at the fourth minute before VF onset [5]. This study followed the conventional approach used in most previous prediction works, analysing ECG signals at specific times. Specifically, a one-minute ECG segment was selected from each record for HRV analysis and classification. Findings from feature significance testing and performance evaluation in this prior study were leveraged here to facilitate multiple comparisons and highlight the limitations of segment-based analysis. A detailed summary of this foundational prior work is presented in Sect. 2.3.

In this study, continuous analyses using the 91-minute signal were implemented to investigate how the features and models, which were obtained using fourth-minute signals, perform over a longer duration. Several features and their corresponding model hyperparameters were selected for model training based on the prediction performance in the previous study. The trained model was used to classify all signals up to 91 minutes before the annotated time in each record using the leave-one-out cross-validation

scheme. This continuous analysis using longer signal was an approach inspired by [14, 15, 19, 20] and represented a more comprehensive approach compared to multi-interval prediction strategies used in studies such as [8–10, 12, 21]. The details of the analysis are provided in Sect. 2.4.

However, the raw SVM outputs from the experiment using longer signal could exhibit inconsistencies due to the non-stationary nature of ECG and HRV signals. To enhance prediction performance, a regularisation method was applied to the outputs of the SVM classifiers. Moving averages with window lengths of 5, 10, and 15 minutes were applied to the SVM outputs at one-minute time steps, generating a final classification output. An alarm was raised if the final output exceeded the threshold of 0.5 at any time step. This method was inspired by the firing power technique used in seizure prediction, which achieved notable prediction results despite its simplicity [22, 23]. The details of the experiment are provided in Sect. 2.5.

2.3 Summary of the previous study on analysis at fourth minute

The previous study [5] employed an analysis approach focused on a specific time point, a method commonly adopted by most prior prediction work. Its methodology is briefly summarised here to highlight the differences between segment-based and continuous analysis. In the previous study, a three-minute prediction window and SVM classifiers were chosen due to their widespread use in the literature at that time and their empirically demonstrated discriminative power [7, 24]. A simplified representation of the workflow is presented in Fig. 1.

Firstly, one-minute signal segments were selected from the fourth minute before VF onset of SDDB and before the two-hour mark of NSRDB (Fig. 3). One segment was selected per record, yielding a total of 38 segments. The same bandpass filter, R peak delineation and HRV signal extraction, as described in Sect. 2.2, were performed on the one-minute signal segments. Eleven time-domain and seven nonlinear features were derived from each signal segment. The features that were occasionally undefined, including sample entropy and long-term scaling exponent, were excluded from further analysis. A two-sample t-test was then applied to the remaining features, and the resulting absolute t-values were used to rank the features for forward feature selection during subsequent classification (see Appendix 3 for detailed statistical results).

Four time-domain features (*rr_std*, *rr_cv*, *rmssd*, *pnn50*) and five nonlinear features (*sd1*, *sd2*, *sd_ratio*, *dfa_alpha1*, *hurst_rs*) were identified as statistically significant and are briefly described below.

1. *rr_std* is standard deviation of RR intervals.

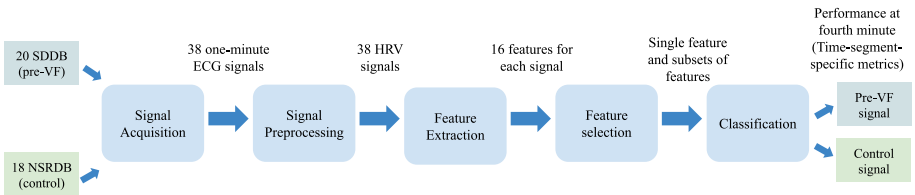


Fig. 1 Simplified workflow of the previous study

2. *rr_cv* is coefficient of variation of RR intervals.
3. *rmssd* is square root of the mean squared differences of successive RR intervals.
4. *pnn50* is proportion of successive RR intervals that differ by more than 50 ms.
5. *sd1* and *sd2* are computed from the fitted ellipse of Poincare plot, where *sd1* quantifies the short-term variability and *sd2* quantifies long-term variability.
6. *sd_ratio* is ratio of *sd1* to *sd2.*
7. *dfa_alpha1* is scaling exponent computed from detrended fluctuation analysis technique, measuring the strength of short-term fractal correlation properties of HRV data in range of 4 to 16 beats.
8. *hurst_rs* (Hurst exponent) measures long-term statistical dependencies and self-similarity in the HRV time series using the rescaled range (RS) method, excluding effects from cyclical patterns [25].

Classification was performed using SVM with various kernel functions, including linear, quadratic, cubic, and Gaussian (radial basis function). The performance of each feature set and classifier configuration was evaluated using a ten-fold cross-validation scheme, repeated 100 times with different random splits. Reported metrics were averaged over these 100 iterations of ten-fold cross-validation to ensure robust performance estimates. The features that achieved the highest average accuracy in time-domain and nonlinear category were *rmssd* (83.32%) and *sd_ratio* (91.87%), respectively. The combination of *hurst_rs* and *sd2* with Gaussian SVM ($\sigma = 1.4$) yielded the highest accuracy of 94.74% among all two-feature classifications. Among several combinations of three features, *hurst_rs*, *sd2* and *pnn50* with Gaussian SVM ($\sigma = 1.7$) achieved the highest average accuracy of 94.66%. For four-feature classification, *hurst_rs*, *sd2*, *pnn50* and *rmssd* with Gaussian SVM ($\sigma = 0.5$) attained an accuracy of 94.61%. Detailed results for the performance evaluation of all feature subsets are provided in Tables 4 and 5.

In the previous study, true positive (TP) represented the number of VF subjects correctly identified using the fourth-minute signal segments while true negative (TN) represented the number of control subjects correctly identified as not having VF at the fourth minute. False negative (FN) reflected the number of actual VF segments misclassified as normal, and false positive (FP) reflected the number of control segments misclassified as VF. Using these four definitions, accuracy, sensitivity and specificity were calculated using Eqs. 1, 2 and 3 to assess the model performance. The performance metrics in the previous study were referred to as time-segment-specific metrics, as they were evaluated on a selected signal segment from each record corresponding to a specific prediction time.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{3}$$

2.4 Continuous analysis using 91-minute signal

This experiment aimed to investigate the algorithm's performance over a longer duration and to compare the results obtained from segment-based analysis with those from continuous analysis. A 91-minute signal was extracted from each record, and the algorithm performance was evaluated using a model trained on the fourth-minute signal segments. SDDB record '37' had the shortest elapsed VF onset time of 01:31:13, making 91 minutes the maximum signal length available to include all SDDB records in the analysis. The feature subsets and model hyperparameters were selected based on the performance result from the previous study. The workflow is summarised in Fig. 2 and is further detailed in the following paragraphs.

ECG signals of 91 minutes each were extracted from the SDDB dataset before the onset of ventricular fibrillation (VF) and from the NSRDB dataset before the two-hour mark. Following extraction, each signal was divided into non-overlapping one-minute segments, resulting in 91 segments per record. Consequently, a total of $38 \times 91 = 3458$ one-minute segments were obtained for subsequent analysis. Again, each one-minute segment underwent the same preprocessing, including bandpass filtering, R peak delineation, and HRV signal extraction. The signal acquisition processes of the previous study and this experiment are juxtaposed in Fig. 3.

HRV features were computed from the extracted HRV signals. The relevant features for extraction were determined based on the prediction performance in the previous study, and their corresponding model hyperparameters were used for subsequent model training and classification. First, the feature subset that reported the highest accuracy, HRV *hurst_rs* and *sd2* with Gaussian SVM ($\sigma = 1.4$), was chosen. To further confirm the research findings, additional comparisons were conducted. To compare the effectiveness of single and multiple features, four subsets including (1) HRV *hurst_rs*, (2) HRV *hurst_rs* and *sd2*, (3) HRV *hurst_rs*, *sd2* and *pnn50* as well as (4) HRV *hurst_rs*, *sd2*, *rmssd* and *pnn50* were chosen for performance evaluation. To evaluate the effectiveness of time-domain versus nonlinear features, two subsets demonstrating the highest performance in each category were selected.

A leave-one-out cross-validation method was employed due to the limited amount of input data. In each iteration, one of the 38 records was designated as the testing record, while the remaining 37 records served as the training set. An SVM model was trained using the fourth-minute signal segments from these 37 training records and the model hyperparameters selected in the previous study. The trained model was then used to classify 91 one-minute signal segments from the testing record. This

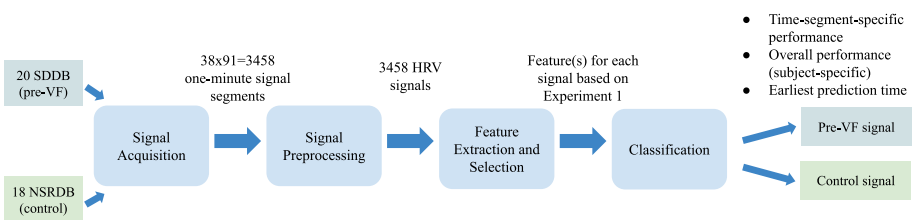


Fig. 2 Simplified workflow of analysis using 91-minute signal

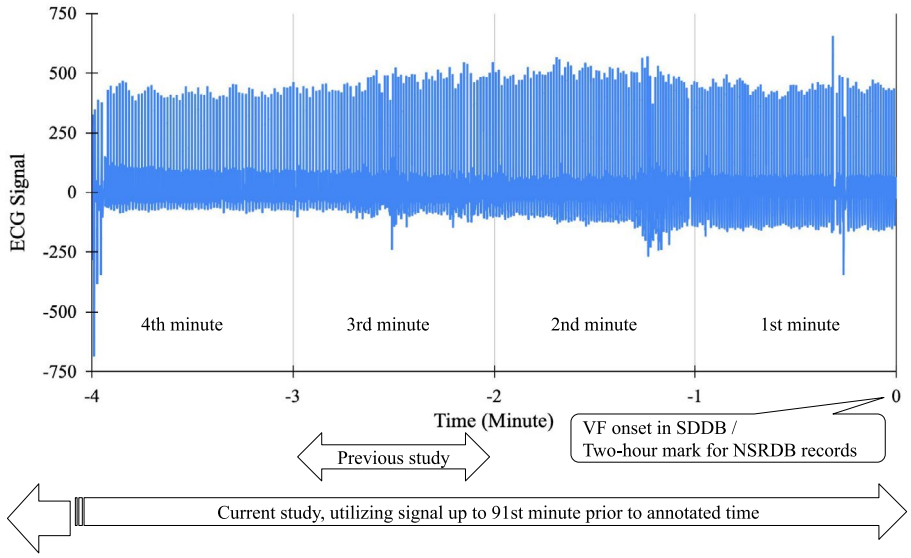


Fig. 3 The previous study used the fourth-minute signal segment for prediction, while this experiment utilised 91 one-minute signal segments from each record for prediction

process was repeated 38 times, with the classification results for the testing record recorded at each step, before calculating the final performance metrics.

In this experiment, additional metrics were calculated, including the earliest prediction time and overall metrics, alongside the time-segment-specific metrics from the previous study. The earliest prediction time was determined by recording the time of the first signal segment in each SDDB record that exhibited a positive result. Both time-segment-specific and overall metrics were calculated using the same Equations (Eqs. 1, 2, 3) to assess the model performance. The time-segment-specific metrics here were similar to those applied in the previous study and most prior studies, but were applied to multiple segments of the longer signal, hence having 91 time-segment-specific metrics for the whole 91-minute signal. The overall metrics can be seen as subject-specific metrics. The overall sensitivity was the probability of correctly predicting imminent VF within the 91-minute signal window of a subject. The overall specificity was the probability of correctly identifying control patient without any positive classifier output throughout the 91 minutes.

A detailed comparison between time-segment-specific and overall metrics is provided in Table 1, highlighting differences in their scope, temporal focus, unit of analysis, and metric assignment rule. As depicted in Fig. 4, the assignment of FP, TN, TP and FN differs between the two approaches. For overall metrics, these classifications are based on whether any one-minute segment within the entire 91-minute signal window is classified as positive. Specifically, records from the NSRDB dataset with any positive classification are considered FP, while those with none are TN; similarly, SDDB records with any positive classification are TP, and those without are FN. In contrast, time-segment-specific metrics evaluate classifier outputs exclusively at the fourth-minute segment. Here, a positive classification at that time point determines TP or FP, while a negative classification corresponds to FN or TN, depending on the

Table 1 Comparison between time-segment-specific metrics and overall metrics

Aspect	Time-segment-specific metrics	Overall metrics
Definition	Metrics calculated for specific signal segments (e.g. fourth minute)	Metrics calculated over the entire pre-event signal window per subject (e.g. 91-minute signal)
Scope	Focuses on classifier performance at predefined time points (typically one segment per subject or multiple fixed segments)	Aggregates classifier outputs across all segments per subject to determine subject-level classification
Temporal focus	Fixed time points	Entire pre-event duration
Unit of analysis	Individual signal segments (e.g. 1-minute segments)	All available segments from record (e.g. 91 segments over 91 minutes)
Metric assignment rule	Based on classifier output at a specific time segment (e.g. fourth minute only)	Based on whether *any* segment in the 91-minute window is classified as positive

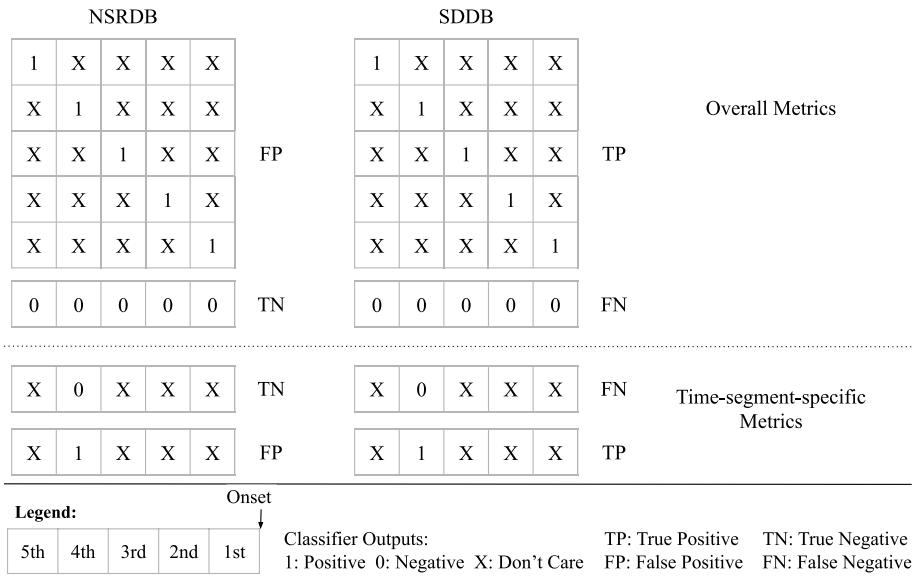


Fig. 4 The different FP, TN, TP and FN definitions of time-segment-specific and overall metrics

dataset. This distinction highlights how overall metrics provide a subject-level assessment across a broader temporal window, whereas time-segment-specific metrics focus on performance at a fixed, predefined moment.

2.5 Regularisation of SVM outputs

The genesis of VF can be viewed as a dynamic process, suggesting that incorporating ECG features with some cumulative measures over time may enhance predictive performance. Furthermore, as a common hard classification problem, misclassification is inevitable and it may lead to frequent false alarms. Regularisation methods have been developed to mitigate this issue, with one such method being the firing power approach which averages several classifier outputs [22]. This method helps smooth out the classifier outputs and reduces the false prediction rate. Building upon the 91-minute continuous analysis, this experiment aimed to improve the algorithm's predictive performance

by generating an alarm only when the averaged SVM outputs exceeded a predetermined threshold.

This experiment introduced an additional step of alarm generation, positioned between classification and performance evaluation. The SVM outputs obtained from the leave-one-out cross-validation in the previous 91-minute analysis were used as input for the alarm generation process. A sliding window of 5, 10 and 15 minutes were moved across the SVM outputs. Within each window, firing power was calculated using Eq. 4 where *fp[n]* was the firing power at the discrete time *n*, *o[k]* was the two-class classifier output at the discrete time *k*, and τ was the window length. The SVM output could either be 1 (pre-VF, positive) or 0 (control, negative). Hence, *fp[n]* of 1 indicated that all signal segments within the window were classified as pre-VF and a zero *fp[n]* indicated that all signal segments within the window were classified as control. The firing power was computed at every minute based on the past τ SVM outputs. An alarm was triggered once the firing power exceeded 0.5. In binary classification, the thresholding of firing power is analogous to the majority voting method, where the alternative (0 or 1) with more than half of the votes is selected.

$$fp[n] = \frac{\sum_{k=n-\tau}^{n} o[k]}{\tau} \tag{4}$$

The earliest prediction time and the overall performance metrics (including accuracy, sensitivity, and specificity) were calculated based on the time series of firing power. The earliest prediction time was defined as the time at which the first alarm was raised for each SDDB record. Overall sensitivity was calculated as the proportion of correctly predicted imminent VF cases, where an alarm was raised within the 91-minute window. Overall specificity was defined as the proportion of correctly identified control patients, where no false alarms were raised.

3 Results

3.1 Continuous analysis using 91-minute signal

We began with the segment-based analysis for comparison with previous studies. The combination of HRV *hurst_rs* and *sd2* with Gaussian SVM ($\sigma = 1.4$) was first chosen for classifier training and testing in this experiment. From the recorded classification outputs, time-segment-specific measures, including accuracy, sensitivity, and specificity, were plotted over time in Fig. 5. The highest accuracy of 94.74% was achieved at the 4th and 57th minutes, followed by 92.11% accuracy at the 22nd and 84th minutes. Maximum sensitivity of 100% was achieved at the 4th and 84th minutes, followed by 95% at the 5th minute. Maximum specificity of 100% was achieved at the 33rd and 57th minutes, followed by 94.44% at several time points: the 3rd, 8th, 9th, 10th, 14th, 15th, 16th, 22nd, 28th, 32nd, 51st, 53rd, 54th, 71st, 73rd, 89th and 90th minutes. As shown in the line chart in Fig. 5, specificity was generally higher than sensitivity. These segment-based results demonstrated high accuracies, aligning with previous studies that predicted VF at multiple specific times [8–10, 12, 21], and this study extended the prediction interval to 91 minutes.

Using the same classification outputs, overall metrics and earliest prediction time were computed. The algorithm achieved 100% sensitivity and an average earliest prediction

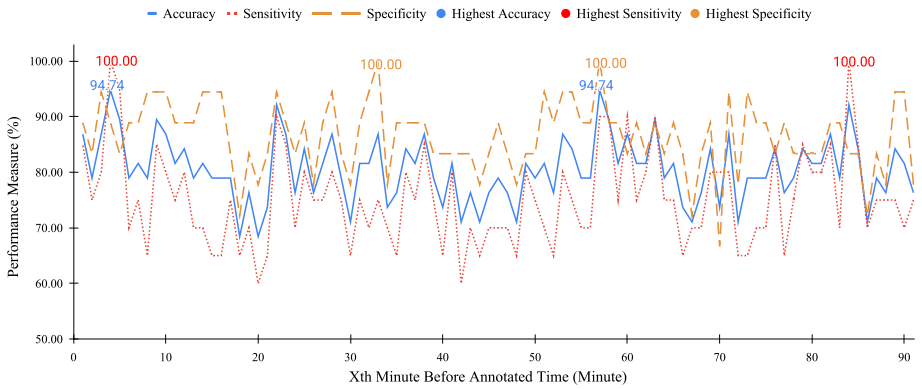


Fig. 5 Time-segment-specific measures over time with the highest accuracy, sensitivity and specificity being annotated

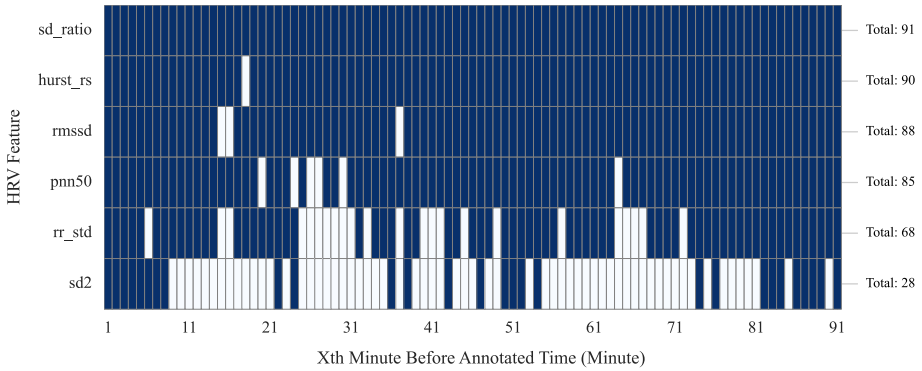


Fig. 6 Binary heat map showing time-resolved statistical significance (1 = $p < 0.05$, 0 = not significant) for each HRV feature over 91 minutes prior to VF. Feature names are shown on the left, and the total number of significant time points for each feature is displayed at the far right

time of 89.05 (±1.96) minutes. However, specificity was low at only 16.67%, leading to false alarms in 15 out of 18 subjects. The overall accuracy was 60.53%. Unlike the segment-based analysis, which generally exhibited higher specificity than sensitivity, the continuous analysis approach produced an overall specificity significantly lower than sensitivity. This finding is consistent with expectations, aligning more closely with prior works that analysed the entire signal continuously using a sliding window and reported average prediction times [14, 15, 19, 20], a methodology that more accurately simulates real-time signal monitoring.

As a complementary analysis, a minute-by-minute statistical significance analysis was conducted across the 91-minute window. Among the selected HRV features, *hurst_rs*, *sd_ratio*, *pnn50*, and *rmssd* were statistically significant ($p < 0.05$) in a large proportion of time points, indicating strong temporal relevance. In contrast, *rr_std* showed intermittent but recurring significance, while *sd2* was significant in much fewer segments overall. These results are visualised in Fig. 6, which presents a binary heat map of significance across the 91-minute interval, with numerical counts indicating how many time points each feature was significant. No feature exhibited consistent significance exclusively in the final phase prior to VF. However, certain features

exhibited localised temporal patterns. For example, *sd2* showed sustained significance near both the beginning (82–91 minutes before VF) and the end (1–8 minutes before VF) of the 91-minute monitoring window.

Additional investigations were conducted on different combinations of features and model hyperparameters. To compare the effectiveness of single and multiple features, four subsets containing one to four features were evaluated for overall performance metrics, as illustrated in Fig. 7. All four classifications achieved the same overall sensitivity of 100% and a similar average earliest prediction time of around 89 minutes. The overall specificity improved incrementally from 5.56% to 16.67% to 27.78%, resulting in corresponding increases in overall accuracy from 50.05% to 60.03% to 65.79% as the number of features increased from one to two to three. However, no improvement occurred when the fourth feature was added, and therefore the three-feature combination was selected for inclusion in the subsequent experiment.

To compare the effectiveness of nonlinear and time-domain features, two subsets of nonlinear features were implemented as a comparison to two subsets of time-domain features. Their performance is presented in Fig. 8. High sensitivity was the highest priority for developing a VF prediction algorithm, thus only the subsets exhibiting 100% sensitivity were considered. HRV *rr_std* and *rmssd* with cubic SVM attained the highest overall accuracy of 68.42% and was selected for inclusion in the subsequent experiment.

Next, a record-specific analysis was conducted using the percentage of positive SVM outputs over the 91 minutes, as depicted in Fig. 9. Each line chart represented a feature subset in combination with its respective SVM kernel type and width. Most models achieved 100% sensitivity (non-zero percentage in the chart), except that *pnn50* and *rmssd* with Gaussian SVM ($\gamma = 1.4$) failed to identify 'sddb/48' as VF. On the other hand, all models achieved specificity below 50%, as indicated by non-zero percentage of positive outputs in more than half of the NSRDB records. Despite the low overall specificity, the percentage of positive SVM outputs was generally higher in SDDB records compared to NSRDB records. The output percentage trends were similar across all models and the result was record-specific. The obvious examples were

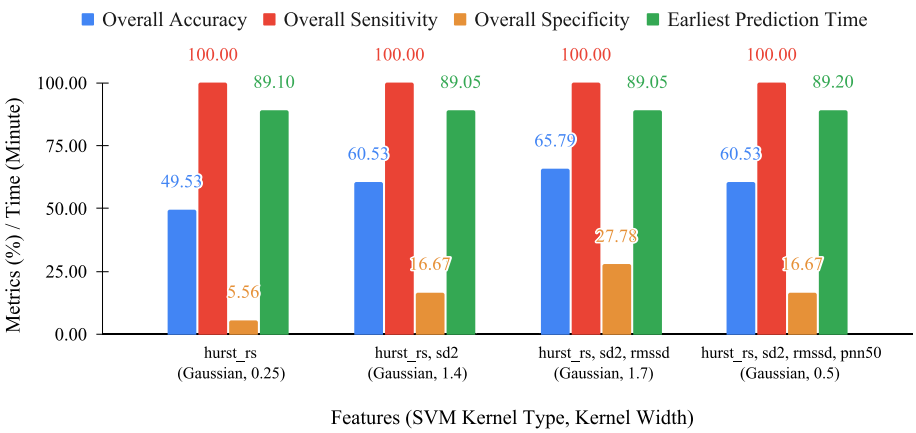


Fig. 7 The overall performance metrics and earliest prediction time of classification using one, two, three and four features

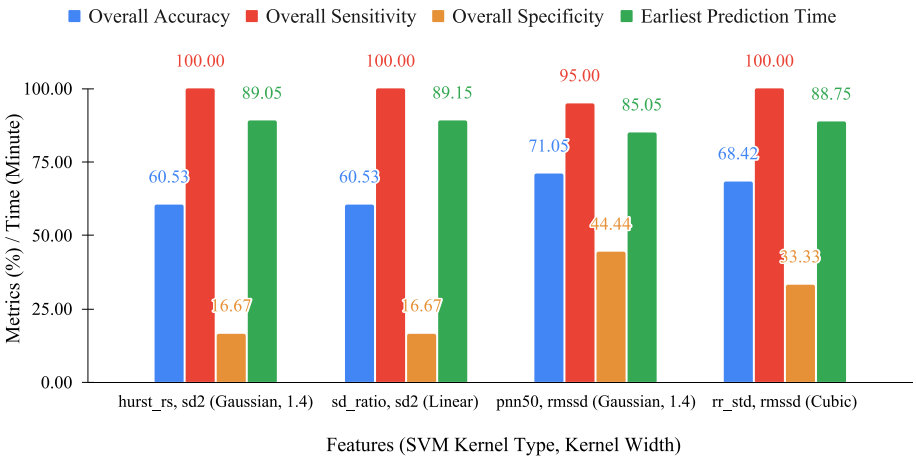


Fig. 8 The overall performance metrics and earliest prediction time of two subsets of nonlinear features (leftmost two) and two subsets of time-domain features (rightmost two)

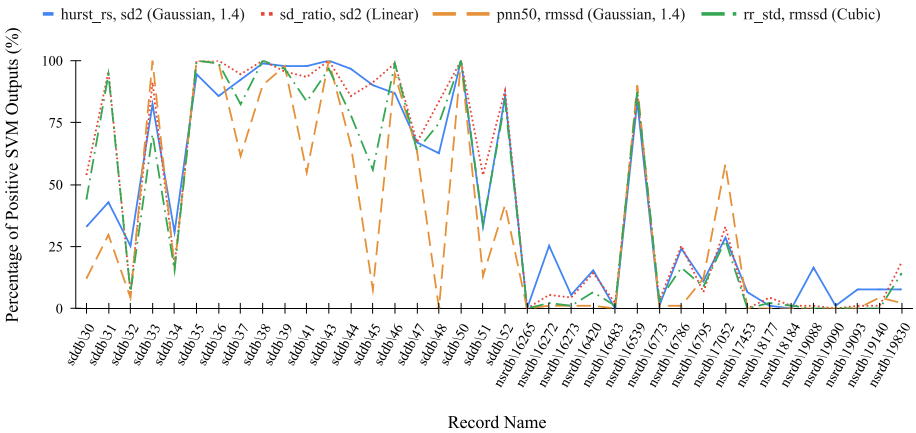
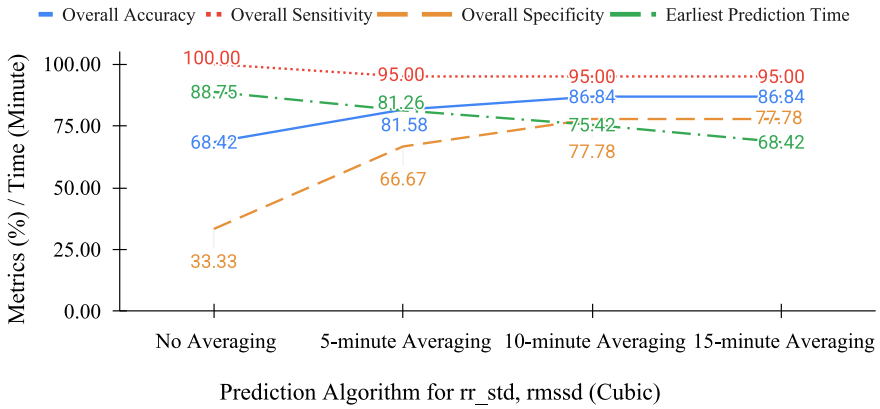


Fig. 9 Percentage of positive SVM outputs over the 91 minutes for each record in SDDB and NSRDB

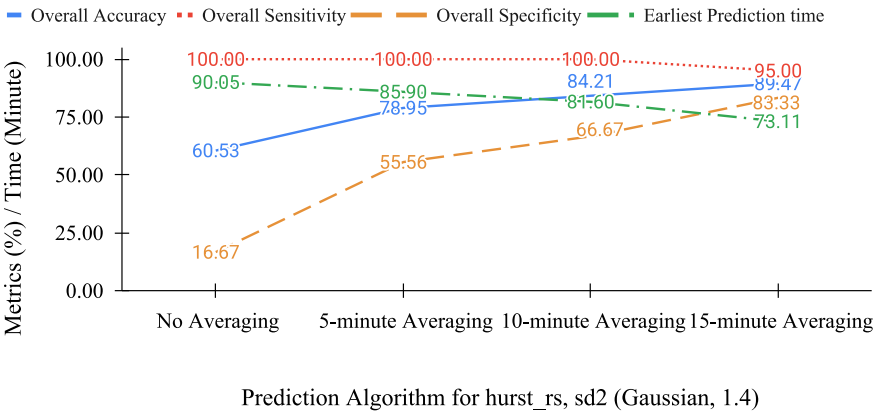
'nsrdb/16539' and 'nsrdb/17052' which consistently showed higher percentages of positive SVM outputs for all models, as well as 'sddb/32' and 'sddb/34' which exhibited lower percentages.

3.2 Regularisation of SVM outputs

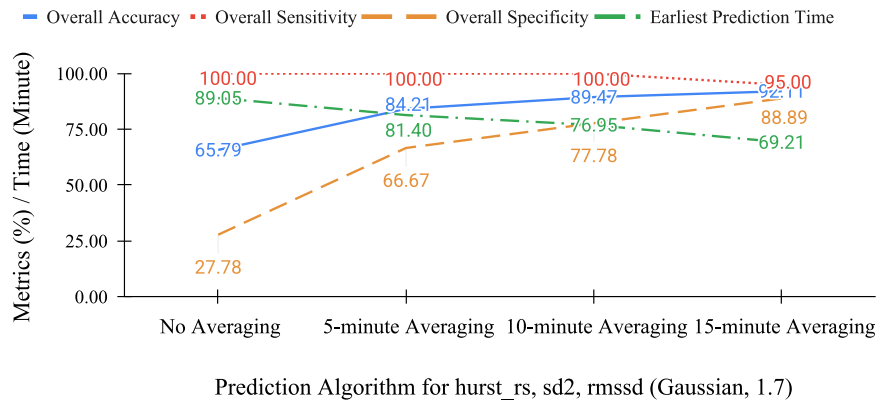
A regularisation method was applied to smooth the SVM classifier outputs using moving average windows of different lengths (5, 10, and 15 minutes), and its impact was evaluated on three-feature subsets with their respective SVM hyperparameters: (*rr_std*, *rmssd*) with cubic SVM; (*hurst_rs*, *sd2*) with Gaussian SVM; and (*hurst_rs*, *sd2*, *rmssd*) with Gaussian SVM. Figure 10 presents the overall performance metrics and earliest prediction times for each configuration, comparing direct SVM outputs against the moving average outputs for the three window lengths. The window sizes (5, 10, 15 minutes) were chosen empirically to evaluate the trade-off between temporal responsiveness and overall prediction performance.



(a)



(b)



(c)

Fig. 10 The overall performance metrics and earliest prediction time achieved using direct SVM outputs and using moving average windows of 5, 10 and 15 minutes, for **a** *rr_std* and *rmssd* with cubic SVM, **b** *hurst_rs* and *sd2* with Gaussian SVM and **c** *hurst_rs*, *sd2* and *rmssd* with Gaussian SVM

Across all three configurations, applying a moving average consistently improved overall specificity and accuracy compared to using direct SVM outputs. However, this came with a slight reduction in sensitivity for longer windows. For example, while both 5-minute and 10-minute windows maintained 100% overall sensitivity, the 15-minute window resulted in a single false negative in all three models, failing to recognise 'sddb/32' as having imminent VF. In addition, the earliest prediction time consistently decreased across all three models as the window length increased, since a longer sequence of positive outputs was required to trigger the alarm. This demonstrates a clear performance trade-off: longer windows are more conservative, enhancing overall specificity and reducing false positives but potentially delaying or missing early warnings. A shorter window retains more temporal sensitivity, allowing faster response to transient changes, but this increased responsiveness comes at the cost of a higher false positive rate.

In terms of model performance, the Gaussian SVM with input features *hurst_rs*, *sd2*, and *rmssd* achieved the highest overall accuracy of 92.1% with a 15-minute window. However, when considering only models that retained 100% sensitivity, the same model with a 10-minute window achieved the highest overall specificity and accuracy at 77.78% and 89.47%, respectively. For this model with 100% overall sensitivity, the earliest prediction times ranged from 11 to 82 minutes before VF onset with a mean of 76.95 ($\pm$15.86) minutes. The minimum earliest prediction time of 11 minutes was recorded in 'sddb/34', followed by 74 minutes in 'sddb/51' and 80 minutes in 'sddb/47'. All other subjects achieved the maximum earliest prediction time of 82 minutes. The three subjects with later prediction times had no documented medical history or medications, with 'sddb/34' and 'sddb/47' being 34-year-old males with sinus rhythm, and 'sddb/51' a 68-year-old female with sinus rhythm and intermittent pacing.

Although the mean prediction time was close to 77 minutes, 'sddb/34' represented a significant outlier, with a prediction time as late as 11 minutes prior to VF onset. Figure 11 visualises the firing power trajectories of 'sddb/34' in comparison with 'sddb/41' (82-minute prediction, unknown age) and 'sddb/47' (80-minute prediction, same age and rhythm profile as 'sddb/34'). The firing power signal of 'sddb/41' remained consistently

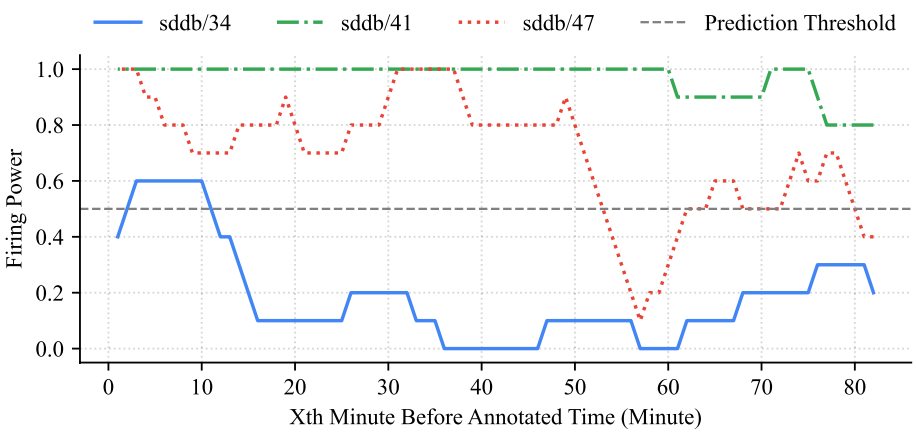


Fig. 11 Firing power trajectories over a 91-minute observation window for three subjects: 'sddb/34' (11-minute prediction), 'sddb/41' (82-minute prediction), and 'sddb/47' (80-minute prediction). The horizontal dashed line at 0.5 indicates the firing power threshold

high, staying above 0.75 throughout the observation window. In contrast, the firing signal in 'sddb/34' remained suppressed below 0.3 until a marked increase began at 16th minute, reaching 0.5 by the 11th minute, thus triggering the alarm. Interestingly, although 'sddb/47' achieved an early prediction time of 80 minutes, its firing power temporarily dropped below 0.5 around the 60-minute mark. Due to missing clinical metadata, deeper patient-specific interpretation was not feasible. However, a feature-based analysis was performed. Compared to 'sddb/41', 'sddb/34' showed elevated *hurst_rs* across most of the intervals. Near the prediction time, 'sddb/34' also exhibited increased *rmssd* and *sd2* values.

3.3 Performance benchmarks

Our results were benchmarked against several state-of-the-art algorithms in the literature, as summarised in Table 2. These studies employed different analysis frameworks, experimental setups, and databases, which introduce inherent challenges for direct comparisons. Nevertheless, Table 2 highlights key performance metrics, including accuracy, sensitivity, specificity, and prediction time, to facilitate an informed comparison.

In this study, we adopted segment-based and continuous analysis frameworks, providing evaluations using time-segment-specific and overall metrics, respectively. With time-segment-specific metrics, the proposed method achieved an accuracy of 94.7%, a sensitivity of 90.0%, a specificity of 100.0%, and a prediction time of 56 minutes. When evaluated on a different time segment, the accuracy, sensitivity, and specificity were 92.1%, 100.0%, and 83.3%, respectively, and the prediction time increased to 83 minutes. Among previous works, [10] reported the longest prediction time of 60 minutes with an accuracy of 92.5%, a sensitivity of 88.33% and a specificity of 100%. In comparison, our

Table 2 Performance benchmark of this study against previous works

Study	Database (number of records)	Accuracy,%	Sensitivity,%	Specificity,%	Prediction time (minutes)
[14]	CUDB (32), PTBDB (32)	98.4	96.9	100.0	4.5 ($\geq$ 0.2)
[15]	SDDB (20), NSRDB (18)	94.7	90.0	100.0	5.4 ($\pm$3.8)
[9]	SDDB (40), NSRDB (36)	97.6	97.9	97.3	13
[26]	SDDB (23), NSRDB (18)	96.3	–	–	25
[27]	SDDB (20), NSRDB (18), CHFDB (15), LTSTDB (51), MITDB (11)	99.3	97.0	99.6	25 (to onset)
[10]	SDDB (20), NSRDB (18)	95.0	93.3	100.0	30
		92.5	88.33	100.0	60
[12]	SDDB (20), NSRDB (18)	100.0	–	–	29
[11]	SDDB (20), NSRDB (18)	99.6	–	–	50
[13]	SDDB (23), NSRDB (18), CHFDB (15)	100.0	100.0	100.0	At least 10
This study (time-segment-specific metrics)	SDDB (20), NSRDB (18)	94.7	90.0	100.0	56
		92.1	100.0	83.3	83
This study (using overall metrics)	SDDB (20), NSRDB (18)	89.5	100.0	77.8	77.0 ($\pm$15.9)

method achieved competitive performance (94.7% and 92.1% accuracies, respectively) at extended prediction times of 56 and 83 minutes, showcasing its superiority over previous studies.

Using overall metrics, our method demonstrated an accuracy of 89.5%, a sensitivity of 100.0%, a specificity of 77.8%, and a mean prediction time of 77.0 minutes with a standard deviation of 15.9 minutes. This earlier prediction time, achieved while maintaining high performance metrics, highlights the additional merits of our approach compared to prior works such as [14, 15], which also employed a continuous analysis framework.

It is worth noting that the methods for reporting prediction times vary across studies. For instance, [12] reported an average accuracy of 97.28% across 30 different prediction times, which is not directly comparable to the overall metrics or time-segment-specific metrics reported in most other studies. To ensure consistency, we report the time-segment-specific metrics from [12] (100% accuracy with 29-minute prediction time) in Table 2.

4 Discussion

This study investigated the performance of selected feature sets and model hyperparameters over a 91-minute interval, identifying key limitations in segment-based evaluation approaches. The variable time-segment-specific metrics, illustrated in Fig. 5, highlighted why previous studies often relied on local subset selection [8] or tested various combinations of features and models [5, 7, 9, 10, 12] at specific prediction times. Notably, this study achieved consistently high accuracies at multiple time points, including the 4th (94.74%), the 57th (94.74%), the 22nd (92.11%), and the 84th (92.11%) minutes, surpassing state-of-the-art benchmarks. However, claiming high accuracy at a single time point, such as the 84th minute, would be misleading, as such evaluations involve short signal segments tied to specific prediction times. This approach does not reflect real-world scenarios with unknown VF onset times and risks inflating specificity. This underscores a critical limitation of time-segment-specific metrics, which fail to comprehensively evaluate algorithms designed for continuous analysis.

To address this limitation, we proposed and employed overall metrics, which evaluate the algorithm's performance using the entire 91-minute signal preceding the annotated VF event. These overall metrics provide a more reliable assessment of an algorithm's prediction capability. Using time-domain features (*rr_std* and *rmssd*) classified using a cubic SVM, the highest overall performance was achieved, including 100% sensitivity, 33.33% specificity, and 68.42% accuracy. Interestingly, these results challenge the dominance of nonlinear features reported in previous studies [5], reinforcing the importance of evaluating algorithms using continuous signal analysis. Furthermore, we propose the term 'average earliest prediction time' to better contextualise the prediction timing. This term acknowledges the variable preparation time required between alarm generation and the actual VF event, providing a more intuitive metric for clinical applications. For instance, an algorithm achieving an average accuracy of 90% with an average prediction time of 60 minutes clearly indicates a 90% likelihood of correctly identifying a patient at risk of a future VF event, with an average of 60 minutes available for clinical intervention. This metric not only enhances the interpretability of algorithm performance but also aligns

with the practical needs of healthcare professionals, who require actionable insights within well-defined time windows.

The percentage of positive SVM outputs over the 91 minutes, as shown in Fig. 9, exhibited similar trends across feature sets, suggesting that the HRV features captured analogous underlying changes during pre-VF progression. Despite the low overall specificity, the higher positive output rate in SDDB versus NSRDB records highlights the potential utility of the firing power method in enhancing performance. To reduce false predictions, we implemented a regularisation method that leveraged the time-accumulative effects of HRV feature changes. By applying a 10-minute sliding window over the classifier outputs of Gaussian SVM models, we classified three HRV features (*hurst_rs*, *sd2*, and *rmssd*) and achieved 100% overall sensitivity, 77.78% overall specificity, and 89.47% overall accuracy. This moving average approach successfully improved the algorithm's specificity without compromising sensitivity by maintaining temporal memory of classifier outputs and suppressing isolated false positives. While this represents a notable improvement, 77.8% specificity may still generate frequent false alarms in continuous monitoring settings, posing a risk of alarm fatigue, where clinicians become desensitised to alerts. Future work should explore adaptive thresholding and advanced regularisation to further reduce false positives, and consider multimodal integration to validate alarms. Minimising false alarms is crucial for ensuring clinical usability and maintaining clinician trust in long-term deployment.

The proposed algorithm is grounded in the physiological relevance of HRV as a surrogate for autonomic nervous system activity, a well-established concept in the analysis of ventricular arrhythmias [28]. We hypothesise that early VF stages involve dynamic shifts in autonomic tone, reflected in distinct HRV feature patterns. Our findings support this hypothesis, showing that *rmssd* tends to be higher and exhibits a broader range in pre-VF signals compared to control signals. This increase in *rmssd* aligns with similar findings reported in [29]. While increased *rmssd* is typically associated with parasympathetic activity, such patterns in pre-VF signals may reflect paradoxical autonomic instability, potentially due to compensatory vagal surges or oscillatory sympathovagal modulation. Similarly, the long-term HRV metric, *sd2*, remained stable and low in controls but exhibited greater variability and occasional high values in pre-VF signals. This increased variance in the pre-VF group may indicate heightened sympathetic influence, potentially increasing arrhythmia susceptibility. These findings partially align with [30], which reported higher *sd2* values in arrhythmic subjects compared to healthy controls.

In contrast, *hurst_rs* is generally lower in pre-VF signals, reflecting reduced signal persistence, with occasional drops below 0.5, suggesting a shift towards anti-persistent or mean-reverting behaviour. This indicates a disruption in the long-range positive correlations typically seen in healthy heart rate dynamics. Reduced *hurst_rs* may point to a loss of complexity and adaptability, which could reflect impaired autonomic regulation and increased myocardial vulnerability. However, nonlinear methods like the Hurst exponent are sensitive to noise and non-stationary data, requiring careful preprocessing. While these methods are not standardised for clinical use, they complement conventional HRV markers by offering insights into heart rate dynamics that linear methods cannot capture. Our results align with [30], where lower Hurst exponent values were reported in arrhythmic patients, and with studies linking reduced complexity to arrhythmogenic

states [28]. Collectively, these findings reinforce the role of vagal modulation, long-term HRV variability and signal complexity in arrhythmogenesis. Notably, subject-specific variability was observed. For instance, 'sddb/34' showed an outlier prediction time of just 11 minutes before VF onset, correlating with elevated *rmssd* and *sd2* near the prediction time, possibly indicating imminent autonomic collapse. These observations highlight the potential value of continuous HRV monitoring for pre-emptive VF prediction.

Furthermore, this study demonstrates the feasibility of early VF prediction, achieving an average earliest prediction time of 76.95 ($\pm$15.86) minutes, while maintaining high accuracy and sensitivity. These findings align with those reported by Fiorina et al., who used a deep learning model with three modalities derived from 24-hour ECG signal to predict arrhythmic risk [31], underscoring the broader potential of early warning systems for life-threatening cardiac events. In contrast to their complex deep learning architecture that integrates multiple ECG-derived modalities, our approach employs a small set of interpretable HRV features and a Gaussian SVM model, offering greater transparency and lower computational complexity. Given their observed performance drop with shorter durations, our method, using one-minute HRV or ECG signals for feature extraction, could complement theirs by enabling granular, real-time predictions suitable for continuous monitoring scenarios. Moreover, this study offers several distinct merits compared to previous work. First, it provides a dual evaluation approach using both time-segment-specific and overall metrics, addressing nuances often overlooked by prior studies focusing solely on segment-specific metrics. Second, it emphasises practical applicability for real-world scenarios, where continuous prediction over extended intervals is critical. Lastly, it introduces a prediction algorithm that achieves, to our knowledge, the earliest prediction time of 77.0 minutes, coupled with a sufficiently high accuracy of 89.5%.

Beyond its predictive performance, the proposed algorithm is designed with future clinical deployment in mind. The algorithm's compatibility with continuous ECG data streams allows seamless integration into existing clinical monitoring infrastructures, such as bedside monitors or centralised telemetry systems. Additionally, its lightweight computational requirements and interpretable output structure make it suitable for implementation in wearable devices or mobile health platforms, broadening its potential to support out-of-hospital early warning applications. From a workflow perspective, the binary alarm system with lead times exceeding one hour offers clinicians a practical window for proactive intervention. Future work will focus on integrating the system with electronic health records and wearable platforms, ensuring compatibility across clinical and remote settings. Evaluating clinician usability and response in real-world environments will help refine the algorithm into a clinically actionable tool for both continuous monitoring and proactive intervention.

Despite encouraging results, the study faced several limitations. First, the comparison between pre-VF segments from patients and control segments from healthy individuals introduces a potential confounding factor: the classifier may partly learn chronic autonomic or structural differences between patients and controls rather than imminent pre-VF deterioration. As such, some of the observed performance may reflect baseline pathology rather than true prodromal signals. Future studies should aim to include control subjects with comparable clinical profiles and incorporate a temporal holdout

design to capture imminent arrhythmogenic changes more effectively. Additionally, feature extraction was restricted to time-domain and nonlinear HRV measures. Incorporating frequency-domain metrics, clinically validated HRV markers and clinical metadata could provide complementary insights for VF prediction.

Second, the limited dataset posed further constraints: the absence of external validation raises concerns about overfitting and limits the generalisability of the results, while the small sample size influenced the modelling strategy. A conventional machine learning pipeline using interpretable HRV features and an SVM classifier was chosen for its robustness in low-data settings. While effective, this approach precludes the use of more expressive deep learning methods such as convolutional neural networks or transformer-based architectures, which may better model nonlinear dependencies. As data availability increases, future work should incorporate external validation, evaluate deep learning methods, and explore earlier prediction windows using longer pre-VF signals. Although our regularisation method enhanced performance, it may not fully address the non-stationary nature of physiological signals. Adaptive thresholding, multi-segment training, and multimodal data integration warrant further investigation. Ultimately, larger-scale, prospective studies remain essential to validate the clinical utility of continuous HRV monitoring for VF prediction.

5 Conclusion

VF remains one of the leading health problems contributing to SCD worldwide, and its timely prediction can significantly enhance survival rates. This study examined both segment-based and continuous analysis frameworks and proposed an improved VF prediction algorithm with high accuracy, early prediction time, and practical clinical applicability. Limitations of segment-based evaluation and time-segment-specific metrics were identified, emphasising the need for subject-based evaluation and overall performance measures for continuous analysis. The 'earliest prediction time' was introduced to replace 'prediction time', reflecting continuous evaluation over extended intervals rather than predictions at discrete time points. The proposed algorithm, utilising HRV features (*hurst_rs*, *sd2* and*rmssd*) with a Gaussian SVM model and a 10-minute sliding window, achieved 100% overall sensitivity, 77.8% overall specificity, 89.5% overall accuracy, and an average earliest prediction time of 77.0 ($\pm$15.9) minutes. These findings demonstrate the feasibility of reliable early VF prediction, offering clinicians sufficient preparation time for intervention.

Appendix A Detailed experimental results of previous experiment

See Tables 3, 4 and 5.

Author contributions

WWH and ELMS conceptualised the research and developed the main methodology. WWH conducted the experiments, performed the data analysis, and wrote the main manuscript text. ELMS and NAAK contributed to system

Table 3 Statistical test results of the previous experiment using features from SDDB and NSRDB

	Mean (SD)		*t*-test Results	
Feature	**NSRDB**	**SDDB**	**Absolute *t*-value**	***p*-value**
rr_counter	81.44 (16.89)	74.15 (28.99)	0.93	0.357
rr_mean	756.09 (151.06)	912.24 (380.89)	1.69	0.103
rr_std	54.61 (34.47)	216.66 (266.11)	2.70	0.014
rr_cv	0.07 (0.04)	0.22 (0.18)	3.60	0.002
rmssd	38.63 (35.58)	275.31 (334.97)	3.14	0.005
pnn50	14.74 (16.79)	44.98 (30.77)	3.81	0.001
pnn20	45.77 (22.72)	60.13 (27.87)	1.73	0.092
tinn_n	673.39 (141.39)	786.19 (301.82)	1.50	0.145
tinn_m	785.41 (162.99)	861.21 (304.38)	0.97	0.340
tinn	112.02 (51.98)	75.02 (80.48)	1.66	0.105
tri_index	8.90 (3.07)	8.71 (5.64)	0.13	0.899
sd1	27.30 (25.15)	194.45 (236.18)	3.15	0.005
sd2	70.96 (42.63)	224.06 (263.25)	2.56	0.018
sd_ratio	2.94 (1.28)	1.23 (0.60)	5.19	0.000
dfa_alpha1	1.26 (0.33)	0.74 (0.30)	5.00	0.000
hurst_rs	0.77 (0.12)	0.47 (0.17)	6.28	0.000

Statistically significant if $p < 0.05$

Table 4 Performance metrics for classifying one and two features using selected model hyperparameters, averaged over 100 iterations of 10-fold cross-validation

Feature category	Feature(s)	Top SVM kernel	Accuracy, %	Sensitivity, %	Specificity, %
1 Time Feature	*rr_std*	Quadratic	76.24 (±0.59)	60.00 (±0.00)	94.28 (±1.24)
	rr_cv	Linear	77.45 (±1.60)	64.25 (±1.79)	92.11 (±2.76)
	rmssd	Linear	83.32 (±1.25)	73.30 (±2.38)	94.44 (±0.00)
	pnn50	Linear	72.18 (±1.51)	55.00 (±0.00)	91.28 (±3.18)
1 Nonlinear Feature	*sd1*	Linear	83.21 (±1.28)	73.10 (±2.44)	94.44 (±0.00)
	sd2	Quadratic	71.63 (±2.50)	53.35 (±2.93)	91.94 (±3.74)
	sd_ratio	Cubic	91.87 (±1.72)	90.00 (±0.00)	93.94 (±3.63)
	dfa_alpha1	Gaussian ($\sigma = 4$)	79.74 (±1.27)	81.50 (±2.41)	77.78 (±0.00)
	hurst_rs	Gaussian ($\sigma = 0.25$)	89.79 (±1.24)	95.00 (±0.00)	84.11 (±2.62)
2 Time Features	*pnn50, rr_std*	Linear	75.42 (±2.83)	59.85 (±4.90)	92.72 (±2.58)
	pnn50, rr_cv	Linear	77.82 (±3.86)	63.85 (±6.77)	93.33 (±2.23)
	pnn50, rmssd	Gaussian ($\sigma = 1.4$)	84.26 (±0.75)	75.10 (±1.42)	94.44 (±0.00)
	rmssd, rr_cv	Cubic	84.11 (±2.92)	83.75 (±5.19)	84.50 (±2.27)
	rmssd, rr_std	Cubic	85.05 (±2.56)	84.15 (±3.63)	86.06 (±3.83)
2 Nonlinear Features	*hurst_rs, sd1*	Gaussian ($\sigma = 1.4$)	94.24 (±1.04)	99.05 (±1.97)	88.89 (±0.00)
	hurst_rs, sd2	Gaussian ($\sigma = 1.4$)	94.74 (±0.00)	100 (±0.00)	88.89 (±0.00)
	hurst_rs, sd_ratio	Gaussian ($\sigma = 1.4$)	87.00 (±1.17)	90.00 (±0.00)	83.67 (±2.47)
	hurst_rs, dfa_alpha1	Gaussian ($\sigma = 1.4$)	89.32 (±0.63)	94.70 (±1.19)	83.33 (±0.00)
	sd_ratio, sd1	Linear	89.42 (±1.50)	89.55 (±2.57)	89.28 (±1.42)
	sd_ratio, sd2	Linear	91.97 (±1.42)	94.10 (±1.93)	89.61 (±1.88)
	sd_ratio, dfa_alpha1	Cubic	89.68 (±2.17)	89.75 (±1.10)	89.61 (±4.15)
1 Time and 1 Nonlinear Features	*pnn50, hurst_rs*	Gaussian ($\sigma = 1.4$)	92.66 (±1.95)	99.00 (±2.13)	85.61 (±3.54)
	rmssd, sd_ratio	Linear	89.63 (±1.53)	89.65 (±2.28)	89.61 (±1.88)
	pnn50, sd_ratio	Linear	87.29 (±1.59)	90.05 (±0.50)	84.22 (±3.32)
	rmssd, hurst_rs	Gaussian ($\sigma = 1.4$)	94.16 (±1.10)	98.90 (±2.08)	88.89 (±0.00)

Table 5 Performance metrics for classifying three and four features using selected model hyperparameters, averaged over 100 iterations of 10-fold cross-validation

Feature category	Feature(s)	Top SVM kernel	Acc, %	Se, %	Sp, %
3 Features	*hurst_rs, sd2, sd1*	Linear	92.32 (±1.38)	95.20 (±2.55)	89.11 (±1.09)
	hurst_rs, sd2, sd_ratio	Linear	91.53 (±1.16)	93.95 (±2.05)	88.83 (±0.97)
	hurst_rs, sd2, dfa_alpha1	Linear	92.05 (±1.59)	94.95 (±2.97)	88.83 (±0.56)
	hurst_rs, sd2, pnn50	Gaussian ($\sigma = 1.7$)	94.55 (±0.67)	99.65 (±1.28)	88.89 (±0.00)
	hurst_rs, sd2, rmssd	Gaussian ($\sigma = 1.7$)	94.66 (±0.45)	99.85 (±0.86)	88.89 (±0.00)
4 Features	*hurst_rs, sd2, pnn50, rmssd*	Gaussian ($\sigma = 0.5$)	94.37 (±1.40)	100.00 (±0.00)	88.11 (±2.96)

Acc, Accuracy; Se, Sensitivity; Sp, Specificity

implementation and jointly supervised the project. XXF coordinated the manuscript submission and provided clinical interpretation of the findings. WHC provided expertise in computational algorithms and contributed to the system's technical validation. ZAA contributed to manuscript refinement and restructuring for clarity and consistency. All authors reviewed and approved the final manuscript.

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Data availability
The electrocardiogram data are available at https://physionet.org/.

Declarations

Conflict of interest
The authors have no conflict of interest to declare that are relevant to the content of this article.

Ethics approval and consent to participate
Not applicable.

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